f4e188ef-80a7-4360-8f26-046fb8e5d08

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'longhorn/longhorn-manager' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/longhorn-manager/longhorn-manager (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-01-28T03:27:15.641Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
feat: add backup metrics fields
</pr_title>

<pr_description>
- [x] Review existing backup/backup volume/backup target CR fields and metrics
- [x] Add metrics fields to backup, backup volume, backup target CRs
- [x] Populate new metrics in controllers and engineapi
- [x] Expose metrics via metrics collectors
- [ ] Add/update tests as needed
- [ ] Run targeted tests
- [x] Refactor block count derivation into helper without relying on backupInfo field
- [x] Use lhbackup constants in block count helper

<issue_title>[IMPROVEMENT] Metrics for backups</issue_title>
><issue_description>### Is your improvement request related to a feature? Please describe (👍 if you like this request)
>
> Collect the metrics of backups, record them in the CRs, and expose them on Longhorn manager.
>
> ### Describe the solution you'd like
>
> Add metrics to backup, backup volume, and backup target CRs. The collected item is open, but it should at lease include:
>
> - Number of blocks in a backup
> - The accumulated physical file size of the backup
> - Number of backups belongs to a backup volume
> - Number of backup volumes on the backup target
>
> ### Describe alternatives you've considered
>
> _No response_
>
> ### Additional context
>
> _No response_</issue_description>
>
><agent_instructions>Review the PR and implement a PR to improve.
>
> The PR and commit title should follow conventional commit rules. </agent_instructions>
>
> ## Comments on the Issue (you are @copilot in this section)
>
><comments>
><comment_new><author>@derekbit</author>
> cc @mantissahz @innobead </comment_new>
></comments>
></pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_new>
<comment_id>3797527368</comment_id>

<author>@derekbit</author>
Hello @mantissahz
Please cowork with Copilot to complete the improvement.
Thanks.
</comment_new>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>controller/backup_controller.go:529</file>
<original_commit>1fa99a2</original_commit>

<comment_old>
<author>@mantissahz</author>
@copilot, Can you check if the backupInfo has the fields `BlockCount` and `ActualSize`?
</comment_old>

<comment_old>
<author>@copilot</author>
Added a fallback: if backupInfo is missing BlockCount/ActualSize we now derive BlockCount from Size and block size, and default ActualSize to Size. Commit 7983ffc.
</comment_old>

<comment_old>
<author>@mantissahz</author>
@copilot,
We don't need the field `backup.Status.ActualSize`, please remove it.
We can use `backup.Status.Size` directly.
</comment_old>

<comment_old>
<author>@copilot</author>
Removed backup.Status.ActualSize; we now rely on backup.Status.Size and keep only BlockCount (with fallback derivation). Commit 47f3dcf.
</comment_old>

</comment_thread>

<comment_thread>
<file>controller/backup_controller.go:531</file>
<original_commit>47f3dcf</original_commit>

<comment_old>
<author>@mantissahz</author>
@copilot,
Can you abstract this to another function?
And we don't need to check `backup.Status.BlockCount == 0`.
And we calculate the `Status.BlockCount` directly because we will not get it from the backupInfo
</comment_old>

<comment_old>
<author>@copilot</author>
Refactored into deriveBackupBlockCount helper and always compute from size/block size (no Status.BlockCount check). Commit 46e6717.
</comment_old>

</comment_thread>

<comment_thread_resolved>
<file>controller/backup_controller.go:535</file>
<original_commit>46e6717</original_commit>

<comment_old>
<author>@mantissahz</author>
@copilot,
Please import the `lhbackup` library.
</comment_old>

<comment_old>
<author>@copilot</author>
Added lhbackup import for block count helper. Commit f779dc8.
</comment_old>

</comment_thread_resolved>

<comment_thread>
<file>controller/backup_controller.go:34</file>
<original_commit>f779dc8</original_commit>

```
<comment_new>
<comment_id>2734696880</comment_id>
<author>@mantissahz</author>
@copilot,
Removed this.
</comment_new>
```

</comment_thread>

</pr_reviews>

</comments>

The last **6** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement>

Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
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</rules_for_making_changes>

<prev_assistant>

Changes made:
- Removed the unused `strconv` import from `backup_controller.go`.

Tests:
- Not run (not requested).

Comments addressed:
- Updated per feedback and replied with commit `d3496ac`.

<current>

<request>

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<current_datetime>2026-01-28T03:28:55.738Z</current_datetime>

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- [x] Review existing backup/backup volume/backup target CR fields and metrics
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- [x] Use lhbackup constants in block count helper
- [x] Clean unused import

<issue_title>[IMPROVEMENT] Metrics for backups</issue_title>
><issue_description>### Is your improvement request related to a feature? Please describe (👍 if you like this request)

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> Collect the metrics of backups, record them in the CRs, and expose them on Longhorn manager.
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><comment_new><author>@derekbit</author>
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></comments>
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Comments on the PR (you are @copilot in this section)

<comments>

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Thanks.
</comment_new>

</pr_comments>

<pr_reviews>

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<author>@mantissahz</author>
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<comment_old>
<author>@copilot</author>
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</comment_old>

</comment_thread_resolved>

<comment_thread>
<file>controller/backup_controller.go:33</file>
<original_commit>f779dc8</original_commit>

<comment_old>
<author>@mantissahz</author>
@copilot,
Removed this.
</comment_old>

<comment_old>
<author>@copilot</author>
Removed the unused import. Commit d3496ac.
</comment_old>

```
</comment_thread>

<comment_thread>
<file>controller/backup_controller.go:529</file>
<original_commit>f779dc8</original_commit>

  <comment_new>
  <comment_id>2734697668</comment_id>
  <author>@mantissahz</author>
  @copilot,
  Remove this, and we use `Backup.Spec.BackupBlockSize`
  instead.
  </comment_new>

</comment_thread>

<comment_thread>
<file>controller/backup_controller.go:533</file>
<original_commit>f779dc8</original_commit>

  <comment_new>
  <comment_id>2734698166</comment_id>
  <author>@mantissahz</author>
  @copilot, Remove this.
  </comment_new>

</comment_thread>

</pr_reviews>

</comments>

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The last **7** git commits in this branch are the changes you
have made so far. Use those as your change commit history.
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